STATISTICS — LEVEL 1: COMPLETE INTERVIEW NOTES

FAANG / MAANG Data Science & AI/ML Interview Preparation

Our learning method: first understand the idea in very simple language → connect it to a real-world situation → understand the reasoning/trick → then learn the formula → then calculate → then explain it like an interviewer.

> This version is deliberately more detailed than the first PDF and includes the mental models, denominator tricks, complement tricks, examples and reasoning patterns we used together.

1. Probability — Start With the World

Probability asks one simple question: **“Out of all possible situations, how often does the thing I care about happen?”** The full collection of possible outcomes is the sample space.

> **Example:** Flip a coin: sample space={Heads,Tails}. P(Heads)=1/2.

> **Example:** Choose a card from a 100-card imaginary deck where 20 are red: P(Red)=20/100=20%.

> **Example:** Product example: 100 users, 40 buy. P(Buy)=40/100=40%.

> **TRICK / BRAIN RULE:** Always first identify the **world** (sample space), then identify the event you want, then count favorable / total.

Formula: P(A)=Number of favorable outcomes / Number of total outcomes (when outcomes are equally likely).

Interview connection: Probability is the foundation for A/B testing, conversion, fraud, recommendation and classification problems.

2. Complement — The “1 Minus” Superpower

Complement means **the opposite**. If an event happens with probability p, it does not happen with probability 1–p.

> **Example:** Notification open rate=20%. Not-open rate=1–0.20=80%.

> **Example:** Fraud rate=2%. Non-fraud rate=1–0.02=98%.

> **TRICK / BRAIN RULE:** When the question says **“at least one”**, immediately ask: **“Would it be easier to calculate none?”** Usually yes. Then use 1–P(none).

Formula: P(not A)=1–P(A).
P(at least one)=1–P(none).

Interview connection: “At least one” appears frequently in interviews because the complement makes a messy sum easy.

3. Independence — Does the First Event Change the Second?

Two events are independent when knowing the first outcome does not change the probability of the second.

> **Example:** Coin flips: first flip does not affect the next flip. P(Head on second)=1/2.

> **Example:** Drawing a card, putting it back, then drawing again: the deck returns to the same state, so probabilities stay the same.

> **Example:** Drawing a card and NOT replacing it changes the sample space, so the second draw depends on the first.

TRICK / BRAIN RULE: Ask: **“After event 1 happens, did my world change?”** If the world returns to the same state, independence is often present.

Formula: P(A and B)=P(A)P(B), for independent A and B.

Interview connection: Independent trials are a key assumption behind the Binomial distribution.

4. Conditional Probability — Shrink the World

Conditional probability looks at probability **inside a restricted world**. The phrase “given that” tells you what world to enter.

Example: 100 customers: 40 bought a product; 30 of those buyers are male. P(Male | Bought)=30/40=75%. We ignore the 60 people who did not buy.

Example: If asked P(Bought | Male), the denominator becomes the total number of males, not the 40 buyers.

TRICK / BRAIN RULE: See the symbol A|B as: **“I am standing inside B; now find A.”** The denominator is the **given group**.

Formula: P(A|B)=P(A and B)/P(B).

Interview connection: This denominator trick is extremely important in interviews, especially for fraud, spam, medical tests and recommendation systems.

5. Bayes — Reverse the Question

Bayes becomes useful when the probability you **want** is the reverse of the probability your system naturally gives you.

Example: Spam detector: you may know “if an email is spam, how often does the detector flag it?” But the business question is often “if it was flagged, how likely is it actually spam?” Those are different probabilities.

Example: Think of 10,000 emails: 100 spam and 9,900 legitimate. If 90% of spam is caught, 90 spam emails are flagged. If 5% of legitimate emails are incorrectly flagged, 495 legitimate emails are flagged. Total flagged=585. So P(Spam | Flagged)=90/585≈15.38%.

TRICK / BRAIN RULE: Build a **100/1,000/10,000 people table** when Bayes feels confusing. It turns probability into counting.

Formula: P(A|B)=P(B|A)P(A)/P(B).

Interview connection: Base rates matter. A detector can have high recall and still have low precision when the positive class is rare.

6. The Denominator Trick — Our Most Important Interview Habit

Many probability mistakes are not calculation mistakes; they are **denominator mistakes**.

Example: “What fraction of flagged emails are spam?” → denominator = all flagged emails.

Example: “What fraction of spam emails were flagged?” → denominator = all spam emails.

Example: “What fraction of buyers are male?” → denominator = all buyers.

Example: “What fraction of males bought?” → denominator = all males.

TRICK / BRAIN RULE: Underline the words after **“given,” “among,” “of the flagged,” “of the buyers,”** etc. That group becomes your denominator.

Formula: Conditional probability is often best understood as: numerator = intersection; denominator = the given population.

Interview connection: Interviewers love this because it tests reasoning, not formula memorization.

7. Random Variables — Turning Random Outcomes Into Numbers

A random variable is simply a rule that gives a number to a random outcome.

Example: X = number of purchases made by a customer today. X can be 0,1,2,3,…

Example: X = delivery time in minutes. X could be 31.2, 31.25, 31.251,…

TRICK / BRAIN RULE: The key question is **“Can I count the possible values or do I measure them on a continuous scale?”**

Formula: Discrete → countable values. Continuous → measurable values.

Interview connection: Counts such as clicks, purchases and defects are common discrete variables; latency, height and time are continuous.

8. Discrete vs Continuous — Count It or Measure It

Discrete variables have separated/countable values. Continuous variables can take values across intervals.

Example: Number of clicks=0,1,2,3 → discrete.

Example: Number of fraud cases=0,1,2,… → discrete.

Example: Latency=100.1 ms, 100.11 ms, 100.111 ms,… → continuous.

Example: Height=170.2 cm, 170.21 cm,… → continuous.

TRICK / BRAIN RULE: Count it = discrete. Measure it = continuous.

Interview connection: Choosing PMF vs PDF starts here.

9. Bernoulli — One Yes/No Trial

Bernoulli is the smallest probability experiment: exactly two outcomes, usually written 1=success and 0=failure.

Example: One user either clicks or does not click.

Example: One transaction is fraud or not fraud.

Example: One coin flip is Head or Tail.

TRICK / BRAIN RULE: Bernoulli = **ONE** yes/no trial. If you have one attempt and two outcomes, think Bernoulli.

Formula: $E[X]=p$; $Var(X)=p(1-p)$.

Interview connection: Conversion/click prediction often starts with Bernoulli outcomes.

10. Binomial — Many Bernoulli Trials + Count Successes

Binomial asks: **“After n independent yes/no trials, how many successes did I get?”**

Example: 10 users each have a 20% chance of clicking. X = number of users who click. X can be 0 through 10.

Example: If n=5 and p=0.2, X=4 means exactly 4 successes out of 5.

TRICK / BRAIN RULE: Remember: **Bernoulli = one trial. Binomial = many trials and count the successes.** Look for fixed n, same p and independent trials.

Formula: $P(X=k)=C(n,k)p^k(1-p)^{(n-k)}$.
$E[X]=np$; $Var(X)=np(1-p)$.

Interview connection: Used for conversion counts, defect counts, successes in experiments and many product metrics.

11. Expected Value — The Long-Run Average

Expected value is the average result you would approach if you repeated the random process many times.

Example: Game: 50% chance of ■200 and 50% chance of ■0. Average earning=0.5×200+0.5×0=■100.

Example: You will not necessarily receive ■100 in one game. You receive either ■200 or ■0. ■100 is the long-run average.

TRICK / BRAIN RULE: Think **“If I play this game 10,000 times, what average will my earnings approach?”**

Formula: $E[X]=\Sigma\, xP(X=x)$.

Interview connection: Expected value is used in decision-making, pricing, risk, experimentation and model evaluation.

12. Variance — How Spread Out Are We?

Variance measures how far values tend to move from the mean, with deviations squared.

Example: Values close to the mean have small squared deviations. Values far away have large squared deviations.

Example: An observation 88 away contributes $88^2=7{,}744$ to a squared-deviation calculation, so extreme values matter a lot.

TRICK / BRAIN RULE: Variance is **spread in squared units**. Squaring also prevents positive and negative deviations from cancelling.

Formula: $Var(X)=E[(X-\mu)^2]$.

Interview connection: Variance is sensitive to outliers, which matters in ML feature analysis and statistical modeling.

13. Standard Deviation — Variance Back to Normal Units

Standard deviation is simply the square root of variance, so it returns spread to the original units.

Example: If variance=16, SD=4.

Example: If mean=100 and SD=4: 96 is 1 SD below, 92 is 2 below, 104 is 1 above, 108 is 2 above.

TRICK / BRAIN RULE: Variance = spread squared. **SD = spread you can actually understand in the original unit.**

Formula: $\sigma=\sqrt{Var(X)}$.

Interview connection: SD is easier to interpret than variance because it has the same unit as the data.

14. Covariance — Do Two Things Move Together?

Covariance asks whether two variables tend to move in the same or opposite directions.

Example: Study time increases and exam score tends to increase → positive covariance.

Example: Price increases and quantity purchased tends to decrease → negative covariance.

Example: If one variable moves up while the other has no systematic movement, covariance can be near zero.

TRICK / BRAIN RULE: Covariance answers **direction**: same direction = positive; opposite direction = negative.

Formula: $Cov(X,Y)=E[(X-\mu X)(Y-\mu Y)]$.

Interview connection: Useful for understanding feature relationships before modeling.

15. Correlation — Covariance With a Standard Scale

Correlation measures the strength and direction of a **linear** relationship and is scaled from –1 to +1.

Example: +1 = perfect positive linear relationship.

Example: –1 = perfect negative linear relationship.

Example: 0 = no linear relationship.

TRICK / BRAIN RULE: Important interview trick: **correlation 0 does not mean "no relationship."** A nonlinear relationship can exist with zero correlation.

Formula: $\rho=Cov(X,Y)/(\sigma X\sigma Y)$.

Interview connection: Correlation is widely used in exploratory data analysis and feature screening.

16. Normal Distribution — The Bell Curve

The Normal distribution is symmetric and bell-shaped. Its center is the mean; for a perfectly Normal distribution, mean=median=mode.

Example: Many measurement-like variables can be approximately Normal under suitable conditions.

Example: The 68–95–99.7 rule: about 68% within ±1 SD, 95% within ±2 SD, and 99.7% within ±3 SD.

TRICK / BRAIN RULE: Picture a person standing at the mean. One SD is one standard step. Two SD is two steps, etc.

Formula: $z=(x-\mu)/\sigma$.

Interview connection: Normal assumptions appear in confidence intervals, hypothesis tests and many statistical models.

17. Z-Score — How Many SD Away?

Z-score converts an observation into a common scale: **"How many standard deviations away from the mean is this?"**

Example: Mean=100, SD=4, x=108 → z=(108–100)/4=2. So 108 is 2 SD above the mean.

Example: x=88 → z=(88–100)/4=–3. So 88 is 3 SD below the mean.

TRICK / BRAIN RULE: Read the sign and magnitude separately: sign tells **direction**; magnitude tells **distance**.

Formula: $z=(x-\mu)/\sigma$.

Interview connection: Useful for anomaly detection, standardization and interpreting Normal-distribution probabilities.

18. Population vs Sample

Population is the full group you care about. Sample is the smaller group you actually observe.

Example: All 1,000,000 customers = population. 5,000 selected customers = sample.

Example: Population mean μ is a parameter. Sample mean x-bar is a statistic.

TRICK / BRAIN RULE: Think **"P = Population/Parameter; S = Sample/Statistic."**

Formula: Population → parameter. Sample → statistic.

Interview connection: Most real-world DS work uses samples to learn about populations.

19. Standard Error — How Much Does My Estimate Bounce?

Standard deviation describes variation of observations. **Standard error describes variation of an estimator across repeated samples.**

Example: Suppose population SD=500 and sample size n=5,000. $SE=500/\sqrt{5000}\approx$■7.07.

Example: If n=20,000, $SE=500/\sqrt{20000}\approx$■3.54. Bigger sample → smaller SE.

TRICK / BRAIN RULE: Do not confuse SD with SE. **SD = people/data vary. SE = your estimate varies across samples.**

Formula: $SE(mean)=\sigma/\sqrt{n}$; if σ is unknown, commonly estimate with $s/\sqrt{n}$.

Interview connection: SE is the bridge to confidence intervals and hypothesis testing.

20. Law of Large Numbers (LLN)

LLN says that as the number of observations grows, the sample average tends toward the true expected value.

Example: Fair coin: 7 heads in 10 flips=70%, but with 10,000 flips the observed proportion will typically be much closer to 50%.

Example: The early results can be noisy; more observations stabilize the average.

TRICK / BRAIN RULE: LLN = **more observations → average approaches truth.**

Interview connection: Useful for understanding why metrics stabilize with more data.

21. CLT — The Sampling-Mean Machine

CLT is about **repeated samples**. Take many samples of size n, calculate each sample mean, and look at the distribution of those means. Under appropriate conditions, it becomes approximately Normal for sufficiently large n.

Example: Imagine 10,000 students. Repeatedly sample 100 students and calculate their average score: 68,71,69,70,… The collection of these averages forms a sampling distribution.

Example: Even if individual student scores are not Normal, the distribution of sample means can become approximately Normal.

Example: Example: $\mu=50$, $\sigma=20$, $n=100$ → $SE=20/10=2$. If $n=400$ → $SE=20/20=1$.

TRICK / BRAIN RULE: The biggest CLT trap: **the original data do not magically become Normal.** The distribution of repeated sample means becomes approximately Normal.

Formula: Mean of sample-mean distribution = μ; $SE=\sigma/\sqrt{n}$.

Interview connection: CLT powers confidence intervals, hypothesis tests and A/B-test inference.

22. LLN vs CLT — Do Not Mix Them

These two are related but answer different questions.

Example: LLN: if I keep collecting observations, does my average move toward the true value? Yes, under suitable conditions.

Example: CLT: if I repeatedly take samples and calculate means, what does the distribution of those means look like? Approximately Normal for sufficiently large samples under suitable conditions.

TRICK / BRAIN RULE: LLN = **WHERE THE AVERAGE GOES.** CLT = **WHAT THE DISTRIBUTION OF SAMPLE MEANS LOOKS LIKE.**

Interview connection: This distinction is a common statistics interview question.

23. Skewness — Look at the Tail

Skewness describes asymmetry. The easiest way to reason about it is to look at the long tail.

Example: Right/positive skew: long tail to the right. Large values pull the mean right, so generally mean > median.

Example: Left/negative skew: long tail to the left. Small values pull the mean left, so generally mean < median.

Example: Your rule from the session: **“The mean follows the tail.”**

TRICK / BRAIN RULE: Do not memorize only “right skew = mean bigger.” Picture the tail pulling the mean toward it.

Interview connection: Income, transaction amounts and many business metrics can be strongly skewed; median can better represent a typical observation.

24. Outliers — Unusual Does Not Mean Wrong

An outlier is an observation unusually far from the rest. It may be an error, a rare legitimate case, or the most important signal in the dataset.

Example: Data: 10,11,12,13,100. Mean=29.2 while median=12. The value 100 pulls the mean strongly.

Example: Fraud detection: an extremely large transaction might be an error—or exactly the fraud signal you need.

TRICK / BRAIN RULE: Never automatically delete an outlier. **Investigate first.** Ask whether it is a data-quality problem, a real rare event, or useful signal.

Formula: Common rule of thumb: $|z|>3$ can flag a potential outlier, but this is not universal.

Interview connection: Outlier handling is important in feature engineering, monitoring and fraud/anomaly detection.

25. Percentiles — Position in the Data

A percentile tells you where a value sits relative to the population.

Example: 90th percentile means approximately 90% of observations are at or below that value.

Example: P50=median.

Example: P95 delivery time=42 minutes means approximately 95% of deliveries take 42 minutes or less.

TRICK / BRAIN RULE: Percentile = **POSITION**. It does not mean “95% probability that this one delivery takes 42 minutes.”

Interview connection: Percentiles such as P50, P90, P95 and P99 are extremely common in product metrics, especially latency.

26. Quartiles & IQR

Quartiles divide ordered data into four parts. Q1=P25, Q2=P50, Q3=P75. IQR measures the middle 50%.

Example: If Q1=10 and Q3=30, IQR=30–10=20.

Example: Using the common 1.5×IQR rule: upper fence=30+1.5×20=60; lower fence=10–1.5×20=–20. A value of 65 is a potential outlier.

TRICK / BRAIN RULE: IQR is robust compared with SD because it focuses on the middle 50% and is less sensitive to extreme values.

Formula: IQR=Q3–Q1; upper fence=Q3+1.5IQR; lower fence=Q1–1.5IQR.

Interview connection: Useful for exploratory analysis and robust outlier detection.

27. PMF — Probability Mass Function

PMF is for **discrete** random variables. It tells the probability of an exact value.

Example: Number of complaints X: P(X=0)=0.50, P(X=1)=0.30, P(X=2)=0.15, P(X=3)=0.05.

Example: P(X=2)=0.15 means exactly 2 complaints has 15% probability.

TRICK / BRAIN RULE: PMF = **DISCRETE + EXACTLY**.

Formula: P(X=x) for a discrete random variable.

Interview connection: Binomial and Bernoulli distributions have PMFs.

28. PDF — Probability Density Function

PDF is for **continuous** random variables. Its height represents density; probability comes from **area over an interval**.

Example: For continuous delivery time, P(X=42)=0 under the usual continuous model. But P(40<X≤42) can be positive.

Example: The PDF height itself can even be greater than 1; what must equal 1 is the total area under the curve.

TRICK / BRAIN RULE: PDF = **CONTINUOUS + DENSITY**. Think “area,” not “height = probability.”

Formula: Probability over interval = area under PDF over that interval.

Interview connection: PDFs are central to Normal and other continuous distributions.

29. CDF — Cumulative Distribution Function

CDF answers: **“What is the probability that X is less than or equal to this value?”**

Example: If F(42)=0.80, then P(X≤42)=80%.

Example: If P95 delivery time=42 minutes, then approximately F(42)=0.95.

Example: CDF works for both discrete and continuous variables.

TRICK / BRAIN RULE: CDF = **BOTH + UP TO**. Imagine filling a bucket from the left: probability accumulates as x increases.

Formula: F(x)=P(X≤x).

Interview connection: Percentiles and latency dashboards are naturally understood through CDFs.

30. PMF vs PDF vs CDF — The Fastest Memory Table

Tool	Used for	Question	Mental picture	Example
PMF	Discrete	Exactly what value?	Individual bars	P(X=4)
PDF	Continuous	How much density here?	Curve + area	P(40<X≤42)
CDF	Both	How much up to here?	Accumulating from left	P(X≤42)

Three-word rule: PMF = EXACTLY • PDF = DENSITY/AREA • CDF = UP TO

31. The Tricks We Used — Interview Shortcut Sheet

“At least one” → Think opposite: 1–P(none).

“Given / among / of flagged / of buyers” → That group is the denominator.

“Did the world change?” → If the first event changes the sample space, independence may fail.

One yes/no → Bernoulli.

Many yes/no + count successes → Binomial.

Expected → Think long-run average.

Variance → Squared spread.

SD → Spread in original units.

Positive covariance → Move together; negative = opposite directions.

Right skew → Tail right → mean usually pulled right → mean > median.

Outlier → Unusual ≠ wrong; investigate.

Percentile → Position, not probability of one individual.

PMF → Discrete + exactly.

PDF → Continuous + density; area gives probability.

CDF → Both + ≤ / up to.

LLN → More observations → average approaches truth.

CLT → Repeated sample means → approximately Normal.

SD vs SE → Data spread vs estimator spread.

32. Worked Questions From Our Session

Question / situation	Reasoning	Answer
Probability / complement	20% of users open a notification. What percent do not open?	1–0.20=0.80=80%.
At least one	Each of 3 users has 20% chance of opening. Probability at least one opens?	None open=(0.80)^3. At least one=1–0.80^3=48.8%.
At least one fraud	Each transaction has 2% fraud probability. For 100 independent transactions, probability of at least one fraud?	No fraud=0.98^100. At least one=1–0.98^100≈86.74%.
Five predictions	Each prediction has 90% chance of no mistake. Probability of at least one mistake in 5?	No mistakes=0.9^5=0.59049. At least one=1–0.9^5≈40.95%
Conditional probability	100 customers, 40 bought, 30 buyers male. P(Male|Bought)?	Restrict world to buyers: 30/40=75%.
Reverse conditional	If asked P(Bought|Male), what changes?	Denominator becomes total males. Do not use 40 unless 4
Spam / Bayes	10,000 emails: 100 spam, 9,900 legitimate. 90% spam caught; 5% legitimate falsely flagged.	TP=90; FN=10; FP=495; TN=9,405; total flagged=585; pre
Expected earning	50% chance of ■200, otherwise ■0.	E[X]=0.5×200+0.5×0=■100.

Variance/SD	Variance=16.	SD=√16=4.
Z-score	Mean=100, SD=4, value=108.	z=(108–100)/4=2 → 2 SD above mean.
Outlier example	10,11,12,13,100.	Mean=29.2; median=12. The 100 pulls mean upward.
IQR	Q1=10, Q3=30.	IQR=20; upper fence=60; lower fence=–20.
CDF	If F(42)=0.80.	P(X≤42)=80%.

33. FAANG / MAANG Interview Questions You Should Be Able to Explain

1. Why is “at least one” often solved using a complement?
2. What is the difference between independent and mutually exclusive events?
3. What does “given B” do to the denominator?
4. Explain Bayes theorem without using a formula.
5. Why can a fraud detector with high recall still have low precision?
6. What is the difference between Bernoulli and Binomial?
7. What does expected value mean in plain English?
8. Why do we square deviations for variance?
9. Why is SD easier to interpret than variance?
10. Can correlation be zero when variables are related?
11. What happens to mean and median in a right-skewed distribution?
12. Why are outliers dangerous for mean and variance?
13. What is the difference between population and sample?
14. Explain LLN vs CLT.
15. Does CLT say the original data become Normal?
16. What does standard error mean?
17. What does the 95th percentile mean?
18. Can the PDF value at a point be greater than 1?
19. Why is $P(X=42)=0$ for a continuous variable?
20. Explain PMF, PDF and CDF in one sentence each.

34. Mini Practice — Try Before Looking at the Key

1. A website conversion rate is 8%. What is the probability a randomly selected visitor does not convert?
2. Ten independent users each have a 5% chance of converting. Write the probability of at least one conversion.
3. In 1,000 transactions, 20 are fraud. A detector catches 18 frauds and falsely flags 49 legitimate transactions. What is precision?
4. One coin is flipped once. Bernoulli or Binomial?
5. A coin is flipped 20 times and X is the number of heads. Bernoulli or Binomial?
6. A game gives ■500 with probability 0.2 and ■0 otherwise. What is expected earning?
7. Mean=50 and SD=5. What is the z-score of 40?
8. Which is usually larger in a right-skewed distribution: mean or median?
9. If sample size increases from 100 to 400, what happens to SE of the mean, all else equal?
10. If $F(200\text{ms})=0.95$, what does that tell you about latency?

35. Practice Answer Key

1. $1-0.08=0.92=92\%$.
2. $1-0.95^{10}$.

3. TP=18; FP=49; precision=18/(18+49)≈26.87%.

4. Bernoulli: one yes/no trial.

5. Binomial: repeated independent Bernoulli trials and count heads.

6. ■100=0.2×500+0.8×0.

7. $z=(40-50)/5=-2$.

8. Mean, because the right tail pulls it right.

9. SE is cut in half because $\sqrt{400}=2\sqrt{100}$; $SE \propto 1/\sqrt{n}$.

10. Approximately 95% of observations are at or below 200 ms.

36. Level 1 Final Mental Map

Probability → how likely?
Complement → opposite → 1–P
Independence → did the first event change the world?
Conditional → shrink the world to “given”
Bayes → reverse/update the probability using evidence
Random variable → turn outcome into a number
Discrete → count
Continuous → measure
Bernoulli → one yes/no
Binomial → many yes/no + count successes
Expected value → long-run average
Variance → squared spread
SD → spread in original units
Covariance → direction together
Correlation → standardized linear relationship
Normal → symmetric bell curve
Z-score → SDs away from mean
Population → whole group
Sample → observed subset
SE → estimator’s bounce across samples
LLN → average approaches truth
CLT → sample means become approximately Normal
Skew → mean follows tail
Outlier → unusual, investigate
Percentile → position
IQR → middle 50% spread
PMF → discrete/exact
PDF → continuous/density/area
CDF → both/up to

37. What We Have Covered vs What Comes Next

Level 1 covered: probability foundations, complements, independence, conditional probability, Bayes, random variables, discrete vs continuous, Bernoulli, Binomial, expected value, variance, SD, covariance, correlation, Normal distribution, z-scores, population/sample, standard error, LLN, CLT, skewness, outliers, percentiles, quartiles, IQR, PMF, PDF and CDF.

Next levels: estimation and confidence intervals → hypothesis testing → p-values and statistical significance → A/B testing → common distributions and sampling → regression/statistical modeling → advanced interview case studies.

Study rule: Before moving on, practice explaining every Level 1 concept in plain English without looking at a formula. Then calculate. Then solve a product-based scenario. That is the interview-ready sequence.